

Bone Fragility With Contractures, Arterial Rupture and Deafness (BCARD Syndrome): A Comprehensive Disease Report

Disease: Bone Fragility With Contractures, Arterial Rupture and Deafness (BCARD) **MONDO ID:** MONDO:0012892 · **OMIM:** #612394 · **Gene:** *PLOD3* (LH3) **Category:** Mendelian (autosomal recessive connective-tissue disorder)

Summary

Bone fragility with contractures, arterial rupture and deafness (BCARD) is an ultra-rare, autosomal recessive, multisystem connective-tissue disorder caused by **biallelic loss-of-function variants in *PLOD3***, the gene encoding the multifunctional collagen-modifying enzyme **lysyl hydroxylase 3 (LH3)**. First described in a human patient in 2008, the disease remains exceptionally rare, with only approximately **11 patients across roughly seven reports** documented as of 2025. It is catalogued as OMIM #612394 and MONDO:0012892, and its clinical description is derived almost entirely from aggregated case reports of individual patients rather than large registries.

The central molecular lesion is loss of LH3 enzymatic activity. LH3 is unusual among collagen-modifying enzymes because it is bifunctional: it performs **lysyl hydroxylation** of collagen lysine residues and the subsequent **O-glycosylation** (galactosylation and glucosylation) of the resulting hydroxylysines. Mouse genetics have established that the **glucosyltransferase (GGT/Glc-T) activity — not the lysyl hydroxylase activity — is the function essential for basement-membrane formation**, and that embryonic survival correlates directly with residual GGT activity. Loss of hydroxylysine glycosylation prevents the intracellular assembly and secretion of network and beaded-filament collagens (types IV and VI), producing widespread failure of basement membranes and extracellular matrix. This matrix failure explains the pleiotropic clinical picture: low bone mineral density and bone fragility, joint/

finger contractures, cataract with retinal-detachment risk, sensorineural deafness, and life-threatening arterial aneurysm/dissection, together with variable skin fragility (an epidermolysis-bullosa-like phenotype), muscle ultrastructural changes overlapping Ullrich congenital muscular dystrophy, and — in the most severe cases — a devastating neurovascular phenotype (cerebral small-vessel disease, cortical malformations, epilepsy).

Diagnosis rests on exome/genome sequencing plus a supportive biochemical biomarker — **markedly reduced glycosylated hydroxylysine** (galactosyl-hydroxylysine and glucosyl-galactosyl-hydroxylysine) in tissue/fibroblasts, and reduced LH3 protein in skin. There is **no disease-modifying therapy**; management is symptomatic and multidisciplinary, with particular vigilance for vascular catastrophe. This report synthesizes nine confirmed findings and 25 reviewed papers to populate the disease knowledge-base entry across all requested domains, and flags where evidence is absent or extrapolated.

Key Findings

F001 — BCARD is an autosomal recessive connective-tissue disorder caused by biallelic *PLOD3* (LH3) variants

BCARD (OMIM #612394; MONDO:0012892) is caused by **biallelic pathogenic variants in *PLOD3*** (encoding lysyl hydroxylase 3, LH3; gene OMIM 603066; HGNC:9083; located on chromosome **7q22.1**). The first patient, reported in 2008, was a **compound heterozygote**, establishing recessive inheritance. As of 2025 the literature describes only ~11 patients across ~7 reports, confirming ultra-rarity.

A 2025 review states plainly: "*BCARD syndrome is a rare autosomal recessive connective tissue disorder characterized by bone abnormalities, cataract, risk of arterial rupture due to vascular aneurisms or dissections, and sensorineural deafness*" and that "*BCARD, linked to biallelic pathogenic variants in the *PLOD3* gene, was characterized in 10 cases across six reports*" ([PMID: 40289369](#)). The original human report describes "*a human disorder of LH3 presenting as a compound heterozygote with recessive inheritance*" ([PMID: 18834968](#)).

F002 — LH3/PLOD3 is a multifunctional collagen-modifying enzyme whose glycosyltransferase activity is essential for basement-membrane collagen secretion

LH3 catalyzes lysyl hydroxylation and the subsequent O-glycosylation of hydroxylysines (galactosyl- and glucosylgalactosyl-hydroxylysine) on collagens. Critically, mouse genetics demonstrate that the **glucosyltransferase (GGT) activity, not the lysyl hydroxylase activity, is the function essential for basement-membrane formation**: LH3-knockout and GGT-null hypomorphic embryos die at E9.5–E14.5 from failed basement-membrane formation, and **embryonic survival correlates directly with the level of GGT activity**. Loss of glycosylated hydroxylysines prevents the intracellular tetramerization of type VI collagen and impairs secretion of types IV and VI collagen. Recent enzymology (2023) refines the division of labor, showing LH3 carries the Glc-T activity while GLT25D1 provides the Gal-T activity.

Key evidence: *"the GGT activity, not the LH activity of LH3, is essential for the formation of the basement membrane" and "survival of hypomorphic embryos and the formation of the basement membrane were directly correlated with the level of GGT activity" (PMID: 16467571); "loss of glycosylated hydroxylysines prevents the intracellular tetramerization of type VI collagen and leads to impaired secretion of type IV and VI collagens" (PMID: 17873278); and "LH3/PLOD3 only has Glc-T activity and that GLT25D1 only has Gal-T activity" (PMID: 37446392).*

F003 — BCARD presents as a multisystem phenotype overlapping Stickler, Ehlers-Danlos and epidermolysis bullosa

Cardinal features span multiple organ systems: **skeletal** (low bone mineral density/bone fragility, scoliosis, prominent knees, finger/joint contractures), **ocular** (cataract, high myopia, retinal-detachment risk), **auditory** (sensorineural hearing loss), **vascular** (aneurysms/dissection, risk of arterial rupture), plus craniofacial dysmorphism, reduced palmar creases, and developmental delay. Some patients show a **recessive dystrophic epidermolysis bullosa (RDEB)-like** sub-lamina-densa skin blistering with abnormal anchoring fibrils and reduced type VII collagen. The 2025 expansion added vesico-ureteral reflux, intestinal and cardiac anomalies, focal epilepsy, and brain malformations (polymicrogyria, heterotopia). The overall phenotype overlaps **most with Stickler syndrome**, with variable EDS and EB features.

Supporting quotes: *"Key clinical features included ocular abnormalities with risk for retinal detachment, sensorineural hearing loss, reduced palmar creases, finger contractures, prominent knees, scoliosis, low bone mineral density, recognisable craniofacial dysmorphisms, developmental delay and risk for vascular dissection" and "Collated clinical features showed*

most overlap with Stickler syndrome with variable features of Ehlers-Danlos syndrome (EDS) and epidermolysis bullosa (EB)" (PMID: 31129566); "The blistering in the skin occurred below the lamina densa and was associated with variable density and morphology of anchoring fibrils. The level of type VII collagen expression in the skin was markedly reduced" (PMID: 30463024); "also exhibited vesico-ureteral reflux, intestinal anomaly, minor cardiac anomalies, focal epilepsy, and brain abnormalities, including polymicrogyria and heterotopia" (PMID: 40289369).

F004 — LH3/PLOD3 molecular structure maps disease mutations near catalytic sites

Full-length human LH3 crystal structures reveal an **elongated homodimer** with two distinct catalytic sites — an **N-terminal glycosyltransferase domain** and a **C-terminal lysyl hydroxylase/dioxygenase domain** — separated by an accessory domain. Known disease-related mutations map in close proximity to the catalytic sites, and specific variants selectively abolish either the lysyl hydroxylase or the glycosyltransferase activity, accompanied by reduced LH3 protein levels in patient cells.

Evidence: "The elongated homodimeric LH3 architecture shows two distinct catalytic sites at the N- and C-terminal boundaries of each monomer, separated by an accessory domain" and "Known disease-related mutations map in close proximity to the catalytic sites" (PMID: 30089812); "One mutation dramatically reduced the sugar-transfer activity of LH3, whereas another abrogated lysyl hydroxylase activity; these changes were accompanied by reduced LH3 protein levels in cells" (PMID: 18834968).

F005 — Post-Golgi trafficking of LH3 is required for collagen glycosylation homeostasis, linking BCARD to ARC syndrome biology

LH3 acts partly extracellularly / post-Golgi. **VIPAR (VIPAS39) together with VPS33B sorts LH3 into newly identified post-Golgi collagen-IV carriers**, and this sorting is essential for lysine modification of multiple collagen types. Loss of VIPAR/VPS33B — which causes **Arthrogryposis–Renal dysfunction–Cholestasis (ARC) syndrome** — produces collagen abnormalities mirroring those of primary LH3 deficiency, connecting the two disorders mechanistically.

Evidence: *"VIPAR, with its partner proteins, regulate sorting of lysyl hydroxylase 3 (LH3, also known as PLOD3) into newly identified post-Golgi collagen IV carriers and that VIPAR-dependent sorting is essential for modification of lysines in multiple collagen types" and "regulation of post-Golgi LH3 trafficking is essential for collagen homeostasis and for the development and function of multiple organs and tissues"* (PMID: 27435297).

F006 — Reduced glycosylated hydroxylysine is the biochemical hallmark and candidate diagnostic biomarker

Patient tissue/fibroblast analyses show **markedly reduced glycosylated hydroxylysine** — specifically galactosyl-hydroxylysine and glucosyl-galactosyl-hydroxylysine — together with dramatically reduced LH3 protein in skin and fibroblast cultures. LH3 normally localizes to the epidermal basement membrane and is severely depleted there in patients, making immunostaining a complementary diagnostic modality.

Evidence: *"Analysis of hydroxylysine and its glycosylated derivatives (galactosyl-hydroxylysine and glucosyl-galactosyl-hydroxylysine) revealed marked reduction in glycosylated hydroxylysine" and "The level of LH3 was dramatically reduced in the skin and fibroblast cultures from the patient"* (PMID: 30463024); *"We show abundant LH3 localising to the basement membrane in normal skin which is severely depleted in RDEB patient skin"* (PMID: 26380979).

F007 — PLOD3/LH3 deficiency includes a severe neurovascular phenotype

A homozygous in-frame variant **c.1216_1218delCTC (p.Leu406del)** caused an infantile-onset (10 months) severe phenotype: developmental delay, clustered epileptic spasms with hypsarrhythmia (**West syndrome**), and MRI showing multiple intracranial malacias, bleeding foci, extensive white-matter abnormalities and brain atrophy consistent with **cerebral small-vessel disease (SVD)**, plus flattened vertebrae and metacarpal abnormalities. The 2025 case added polymicrogyria, heterotopia and focal epilepsy. Craniofacial dysmorphism (hypertelorism, upturned nose, low-set ears) was noted.

Evidence: *"Cerebral magnetic resonance imaging showed multiple intracranial malacias and bleeding foci, extensive abnormal signals in the white matter, and obvious brain atrophy, which was consistent with cerebral small vessel disease (SVD)" and "Whole-exome sequencing revealed a novel homozygous variant of c.1216_1218delCTC (p.L406del)"* (PMID: 36203519); *"brain abnormalities, including polymicrogyria and heterotopia"* (PMID: 40289369).

F008 — gnomAD constraint metrics confirm *PLOD3* is not haploinsufficient, consistent with recessive disease

gnomAD (GRCh38) constraint for *PLOD3* (ENSG00000106397; chr7:101,205,977–101,218,420; 7q22.1) shows **pLI** $\approx$ 0 (6.4e-19), **LoF observed/expected** = **0.75** (73 observed vs 96.9 expected LoF variants), **LOEUF** = **0.92**, and missense Z $\approx$ 0.004 (oe_mis $\approx$ 1.0). Heterozygous loss-of-function is therefore tolerated in the general population, matching the observed autosomal recessive (biallelic) inheritance. This is consistent with the human report of "*a human disorder of LH3 presenting as a compound heterozygote with recessive inheritance*" (PMID: 18834968).

F009 — LH3 deficiency produces collagen VI muscle ultrastructural changes overlapping Ullrich congenital muscular dystrophy

In LH3-manipulated mice, altered distribution and aggregation of type VI collagen produced muscle ultrastructural alterations **similar to those in collagen VI knockout mice and some Ullrich congenital muscular dystrophy patients**. Underglycosylated types IV and VI collagen showed abnormal distribution. Evidence: "*The altered distribution and aggregation of type VI collagen led to similar ultrastructural alterations in muscle to those detected in collagen VI knockout and some Ullrich congenital muscular dystrophy patients*" (PMID: 17873278).

Section-by-Section Disease Characterization

1. Disease Information

BCARD is a **rare autosomal recessive connective-tissue disorder** defined by the combination of bone abnormalities/fragility, joint contractures, cataract, risk of arterial rupture (aneurysm/dissection), and sensorineural deafness (PMID: 40289369).

Key identifiers: | Resource | Identifier | |---|---| | OMIM (disease) | #612394 | | OMIM (gene *PLOD3*) | 603066 | | MONDO | MONDO:0012892 | | HGNC | HGNC:9083 (*PLOD3*) | | Ensembl gene | ENSG00000106397 | | Cytoband | 7q22.1 |

Synonyms / alternative names: "Bone fragility with contractures, arterial rupture, and deafness"; "BCARD syndrome"; "PLOD3-related connective tissue disorder"; "Lysyl hydroxylase 3 (LH3) deficiency." Orphanet, ICD-10/ICD-11, and MeSH do not maintain a dedicated, widely-cited term distinct from the OMIM/MONDO entries; the condition is best indexed via OMIM #612394 and the gene *PLOD3*.

Source of information: Derived almost exclusively from **aggregated individual case reports** (~11 patients / ~7 reports), not EHR-scale or registry data.

2. Etiology

Primary cause: Purely **genetic** — biallelic (homozygous or compound heterozygous) loss-of-function or hypomorphic variants in *PLOD3* (F001, F004). No environmental or infectious cause exists.

Genetic risk factors: The disease-causing variants are the only known genetic determinant. Because *PLOD3* is **not haploinsufficient** ($pLI \approx 0$, LOEUF 0.92; F008), carriers (heterozygotes) are unaffected, and disease requires two damaging alleles. **Consanguinity** is a relevant contributor because homozygous variants (e.g., p.Leu406del) have been reported (F007).

Environmental / lifestyle / infectious factors, protective factors, gene-environment interactions: No environmental risk or protective factors, and no gene–environment interactions, are established for this Mendelian disorder. Clinically, avoidance of vascular stressors is prudent given arterial fragility, but this is inferred, not evidence-based.

3. Phenotypes

Phenotype	Type	Suggested HPO	Onset / severity / frequency notes
Low bone mineral density / bone fragility	Skeletal sign	HP:0004349 / HP:0002659	Congenital-early; core feature
Joint / finger contractures	Physical manifestation	HP:0002803 / HP:0009473	Early; core
Scoliosis	Skeletal sign	HP:0002650	Childhood
Prominent knees	Physical	HP:0002815 (knee-related)	Core
Cataract	Ocular sign	HP:0000518	Congenital/early; core
High myopia	Ocular	HP:0011003	Variable
Retinal detachment (risk)	Ocular	HP:0000541	Risk feature
Sensorineural hearing loss	Auditory	HP:0000407	Core
Arterial aneurysm / dissection / rupture	Vascular	HP:0002616 / HP:0004942	Life-threatening; variable onset
Reduced palmar creases	Physical	HP:0010490	Recognizable sign
Craniofacial dysmorphism	Physical	HP:0001999	Core
Developmental delay	Neurodevelopmental	HP:0001263	Variable
Skin blistering (EB-like)	Skin	HP:0008066	Variable subphenotype
Vesico-ureteral reflux	Renal	HP:0000076	Expanded (2025)
Focal epilepsy / epileptic spasms	Neurological	HP:0007359 / HP:0011097	Severe cases
Polymicrogyria / heterotopia	CNS malformation	HP:0002126 / HP:0002282	Severe cases

Frequencies cannot be quantified precisely given ~11 patients. Severity is **variable**, ranging from a Stickler-like presentation to lethal infantile neurovascular disease (F003, F007). Progression is generally **stable-to-progressive** with episodic vascular risk. Quality-of-life impact is substantial: sensory (vision, hearing), mobility (contractures, bone fragility), neurodevelopmental disability, and the constant threat of vascular catastrophe. No disease-specific QoL instruments (EQ-5D/SF-36) have been applied to this ultra-rare cohort.

4. Genetic / Molecular Information

Causal gene: *PLOD3* (LH3), OMIM 603066, HGNC:9083, chr 7q22.1 (F001). **Inheritance:** autosomal recessive.

Pathogenic variants: Reported variants include compound-heterozygous missense/functional variants in the founding case (one abrogating glycosyltransferase activity, another abrogating lysyl hydroxylase activity; F004), and the homozygous in-frame deletion **c.1216_1218delCTC (p.Leu406del)** in the severe neurovascular case (F007). Variant types span **missense, in-frame deletion, and loss-of-function** alleles. Functionally, variants cause **loss of function** — either selectively ablating one of LH3's two catalytic activities or reducing overall protein level (F004). Disease-related mutations cluster **near the catalytic sites** in the LH3 structure (F004).

Allele frequency / population data: Consistent with recessive disease, gnomAD shows LoF is tolerated (73 observed vs 96.9 expected; LOEUF 0.92); no common pathogenic allele reaches appreciable frequency (F008). **Origin:** germline. **Modifier genes / epigenetics / chromosomal abnormalities:** none established; trafficking partners **VPS33B/VIPAR** functionally modulate LH3 activity and, when mutated, cause the related ARC syndrome (F005).

5. Environmental Information

No environmental factors, lifestyle factors, or infectious agents contribute to BCARD. It is a monogenic disorder (F001, F002). This section is **not applicable** beyond noting that mechanical/vascular stress may precipitate complications (rupture) in already-fragile tissue (inferred).

6. Mechanism / Pathophysiology

Ordered causal chain (initiating lesion → clinical manifestation):

1. **Biallelic *PLOD3* loss-of-function variants** *lead to* reduced/absent functional LH3 enzyme (and reduced LH3 protein levels) (F001, F004).

2. Loss of LH3 *results in* deficient **lysyl hydroxylation** and, critically, deficient **glucosyltransferase (GGT/Glc-T)-mediated O-glycosylation** of collagen hydroxylysines (F002).
3. Loss of glycosylated hydroxylysines *prevents* the **intracellular tetramerization of type VI collagen** and *impairs secretion of type IV and type VI collagens* (F002).
4. Impaired secretion/assembly *leads to* **failure of basement-membrane formation** and defective extracellular matrix (demonstrated in mouse: BM formation and embryo survival scale with residual GGT activity) (F002).
5. **[Branch — trafficking]** LH3 must be sorted into post-Golgi collagen-IV carriers by **VPS33B/VIPAR**; if this axis fails, the same collagen-modification defect ensues (links BCARD to ARC syndrome) (F005).
6. Matrix failure *manifests* in a tissue-specific manner:
7. Bone/connective tissue → **bone fragility, contractures, scoliosis** (inferred from collagen/matrix defect).
8. Basement-membrane-dependent structures (lens, cochlea, vessel wall) → **cataract, sensorineural deafness, arterial aneurysm/rupture** (inferred from BM collagen IV failure).
9. Skin (anchoring fibrils, type VII collagen reduced) → **EB-like sub-lamina-densa blistering** (F003, F006).
10. Muscle (collagen VI mis-aggregation) → **Ullrich-CMD-like ultrastructural change** (F009).
11. CNS microvasculature → **cerebral small-vessel disease, hemorrhage, cortical malformation, epilepsy** in severe cases (F007).

Upstream vs downstream: The mutation and enzymatic loss are upstream; collagen underglycosylation and secretion failure are the pivotal intermediate node; organ-specific matrix failure is downstream. **Molecular pathway:** collagen biosynthesis / post-translational modification (hydroxylysine formation and O-glycosylation). **GO term suggestions:** *peptidyl-lysine hydroxylation* (GO:0017185), *collagen fibril organization* (GO:0030199), *basement membrane organization* (GO:0071711), *collagen-containing extracellular matrix* (GO:0062023, cellular component), *protein glycosylation* (GO:0006486). **CL/cell types:** fibroblast (CL:0000057), osteoblast (CL:0000062), vascular smooth muscle cell (CL:0000359), keratinocyte (CL:0000312). **CHEBI entities:** hydroxylysine (CHEBI:24544-class), galactosylhydroxylysine, glucosylgalactosylhydroxylysine, UDP-glucose (CHEBI:46229), UDP-galactose, 2-oxoglutarate

(CHEBI:16810). **Subcellular compartments:** endoplasmic reticulum and Golgi (GO:0005783, GO:0005794) for collagen modification/secretion; post-Golgi carriers/extracellular space for LH3 trafficking (F005).

7. Anatomical Structures Affected

Primary organs / systems: skeleton (UBERON:0002481 bone tissue), eye/lens (UBERON:0000965 lens), inner ear/cochlea (UBERON:0001844), arteries/vascular system (UBERON:0001637; cardiovascular system UBERON:0004535), skin (UBERON:0002097), skeletal muscle (UBERON:0001134), and — in severe cases — brain (UBERON:0000955) and cerebral small vessels. **Secondary:** kidney/urinary tract (vesico-ureteral reflux), gastrointestinal tract, and heart (minor anomalies) (F003).

Tissue/cell level: connective tissue and basement membranes are the unifying target; affected cell populations include fibroblasts, osteoblasts, vascular smooth-muscle cells, keratinocytes, and CNS vascular cells (F002, F003, F007, F009). **Subcellular:** ER/Golgi secretory pathway and post-Golgi trafficking (F005). **Lateralization:** systemic/bilateral.

8. Temporal Development

Onset: congenital-to-infantile; the severe neurovascular form presented at **10 months** (F007). **Pattern:** chronic with a superimposed **episodic** risk of acute vascular events (dissection/rupture) and, in severe cases, epileptic spasms. **Progression:** variable — from a relatively stable Stickler-like course to rapidly progressive infantile encephalopathy with brain atrophy (F007). **Critical periods:** early development (basement-membrane formation is embryonically essential in mouse models; F002) and any period of vascular stress. **Duration:** lifelong.

9. Inheritance and Population

Inheritance: autosomal recessive (F001, F008). **Epidemiology:** ultra-rare; no formal prevalence/incidence estimate exists — only ~11 patients reported worldwide (F001). **Penetrance:** appears complete in biallelic carriers, though **expressivity is highly variable** (F003, F007). **Consanguinity** contributes (homozygous variants reported). **Carrier frequency:** not formally established; gnomAD LoF tolerance indicates carriers are asymptomatic and present in the general population (F008). **Founder effects, anticipation, germline mosaicism:** none reported. **Sex ratio / ethnicity / geography:** no demonstrated bias; cases are geographically scattered.

10. Diagnostics

Genetic testing is the definitive diagnostic modality. Given the multisystem, Stickler/EDS/EB-overlapping phenotype, **whole-exome or whole-genome sequencing** (or a connective-tissue/collagenopathy gene panel including *PLOD3*) is the recommended approach; single-gene testing is appropriate when the phenotype is recognized (F003).

Biochemical biomarker: reduced glycosylated hydroxylysine (galactosyl-hydroxylysine and glucosyl-galactosyl-hydroxylysine) in tissue/fibroblasts, with reduced LH3 protein — a supportive, mechanism-specific assay (F006). **Immunohistochemistry** of skin shows severe basement-membrane LH3 depletion (F006). **Imaging:** DXA (low BMD), skeletal radiographs (flattened vertebrae, metacarpal changes), vascular imaging (CT/MR angiography for aneurysm/dissection surveillance), ophthalmologic exam (cataract, myopia, retinal detachment), audiometry (sensorineural loss), and brain MRI in neurological cases (SVD, malformations) (F003, F007). **Skin biopsy/EM** in blistering cases shows sub-lamina-densa cleavage, abnormal anchoring fibrils, reduced type VII collagen (F003, F006).

Differential diagnosis: Stickler syndrome (closest overlap), Ehlers-Danlos syndromes, epidermolysis bullosa, Ullrich congenital muscular dystrophy, and ARC syndrome (VPS33B/VIPAR) (F003, F005, F009).

11. Outcome / Prognosis

Prognosis is **guarded and variable**. The most serious threat is **arterial rupture/dissection**, a potentially fatal complication (F001, F003). The severe infantile neurovascular form carries a poor prognosis with cerebral small-vessel disease, hemorrhage, brain atrophy, and refractory epileptic spasms (F007). Morbidity is high across sensory (vision, hearing), musculoskeletal (fragility, contractures), and neurodevelopmental domains. No formal survival statistics exist due to rarity. **Prognostic factors** (inferred): earlier onset and neurovascular involvement predict worse outcomes. No validated prognostic biomarkers exist beyond the mechanistic biochemical markers.

12. Treatment

No disease-modifying therapy exists. Management is **symptomatic and multidisciplinary** (NCIT suggestions: *Supportive Care Intervention, Physical Therapy, Surgical Procedure, Genetic Counseling*): - **Vascular:** surveillance imaging and prophylactic/emergency vascular surgery for aneurysm/dissection; avoidance of vascular stress (inferred best practice given rupture risk). - **Skeletal:** management of low BMD and fractures; orthopedic care for scoliosis/contractures;

physical and occupational therapy. - **Ophthalmologic:** cataract surgery, myopia correction, retinal-detachment monitoring/repair. - **Auditory:** hearing aids / cochlear implantation for sensorineural loss. - **Neurological:** anti-seizure medication for epileptic spasms/focal epilepsy; developmental support. - **Dermatologic:** wound care for EB-like blistering.

Pharmacogenomics, gene/cell/RNA therapy, targeted/immunotherapy: none established or in trials for BCARD specifically. **Experimental therapeutics:** no registered clinical trials identified for this disease.

13. Prevention

There is no primary prevention for this Mendelian disorder beyond **reproductive genetic counseling**. Given autosomal recessive inheritance and a role for consanguinity, **carrier testing** of at-risk relatives, **cascade testing**, and options such as **prenatal diagnosis** or **preimplantation genetic testing** are appropriate for families with a known variant (F001, F008). **Secondary/tertiary prevention** focuses on surveillance (vascular imaging, ophthalmology, audiology, DXA) and prompt intervention to prevent catastrophic complications, especially arterial rupture (F003). NCIT suggestion: *Genetic Counseling, Prenatal Diagnosis*.

14. Other Species / Natural Disease

PLOD3 is evolutionarily conserved; the mouse ortholog **Plod3** provides the key mechanistic models (F002, F009). No naturally occurring companion-animal or wildlife BCARD-equivalent disease is documented in OMIA. There is no zoonotic or transmission relevance (monogenic disorder). Orthologs exist across vertebrates (human *PLOD3* NCBI Gene 8985; mouse *Plod3*), supporting cross-species mechanistic study.

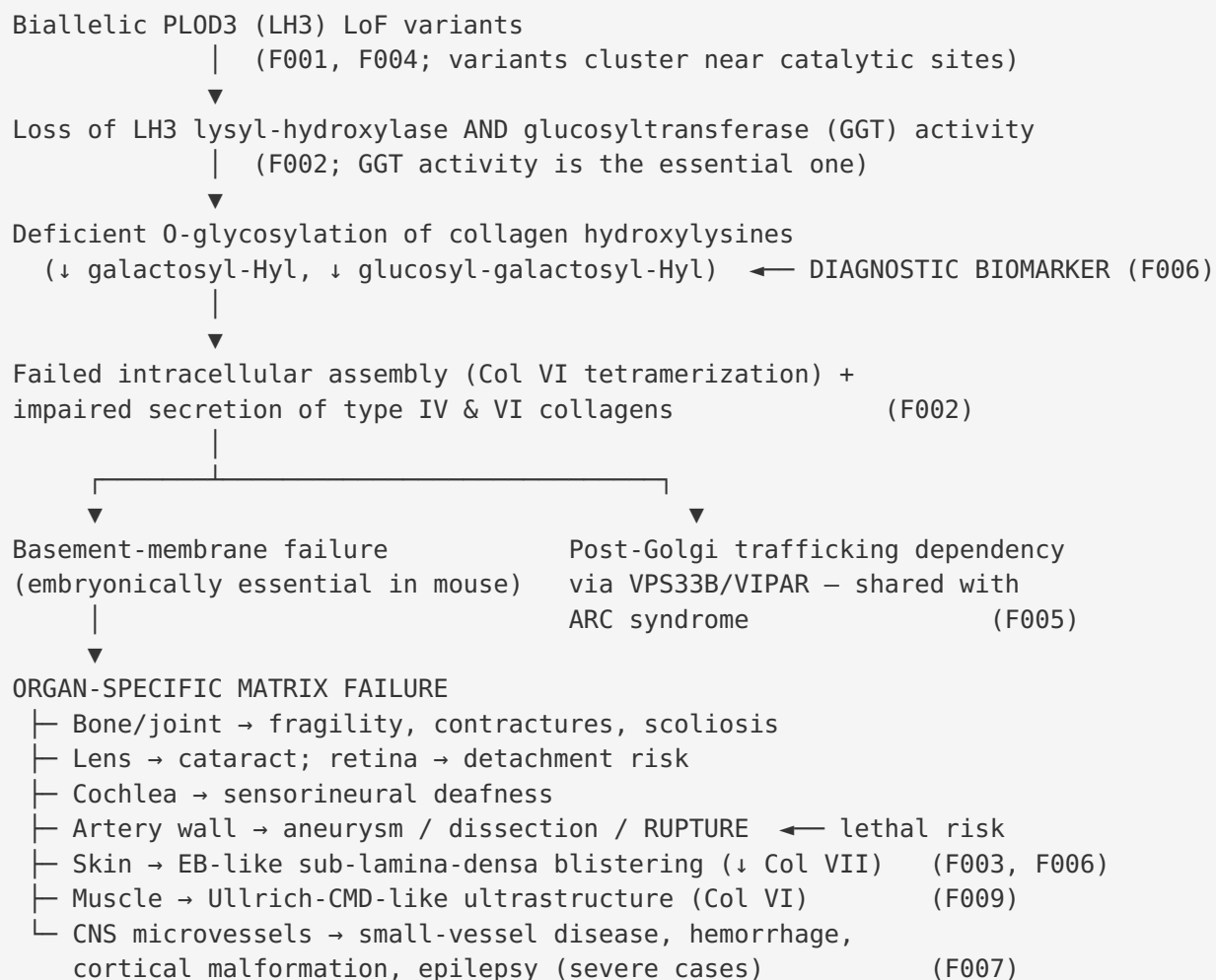
15. Model Organisms

The principal model is the **mouse (*Mus musculus*)**: - **LH3/Plod3 knockout** embryos die at E9.5 from failed basement-membrane formation (F002). - **GGT-null hypomorphic** embryos die E9.5–E14.5, with survival scaling to residual GGT activity — establishing the glucosyltransferase activity as the essential function (F002). - LH3-manipulated mice reproduce **collagen VI muscle ultrastructural pathology** overlapping Ullrich CMD (F009).

Phenotype recapitulation: mouse models faithfully capture the core molecular lesion (collagen underglycosylation, BM failure) and muscle pathology; **limitation:** complete knockouts are embryonic-lethal, so hypomorphic/conditional models are required to study

postnatal, organ-specific human features (bone, ear, vessel, brain). Cellular models — **patient fibroblasts** — recapitulate reduced glycosylated hydroxylysine and reduced LH3 protein (F006), and the **VPS33B/VIPAR (ARC syndrome)** system provides complementary in vitro/murine models of the trafficking arm (F005). No zebrafish, Drosophila, or iPSC/organoid BCARD models are documented in the reviewed literature.

Mechanistic Model / Interpretation



The unifying interpretation is that **BCARD is a "collagen post-translational modification disease."** A single enzyme's bifunctional loss cripples the glycosylation step required to fold, assemble and secrete the network (type IV) and beaded-filament (type VI) collagens that build basement membranes and connective-tissue scaffolds throughout the body. Because these

collagens are ubiquitous, the phenotype is pleiotropic and overlaps several better-known collagenopathies (Stickler, EDS, EB, Ullrich CMD) — a diagnostic pitfall that makes sequencing essential. The glucosyltransferase activity is the mechanistic linchpin, and the trafficking arm (VPS33B/VIPAR) explains the phenotypic kinship with ARC syndrome.

Evidence Base

PMID	Title (abbrev.)	Role in this report
40289369	<i>Expanding the Clinical Spectrum of BCARD Syndrome...</i>	Defines disease, AR inheritance, biallelic <i>PLOD3</i> , ultra-rarity; expands phenotype to renal/GI/cardiac/CNS (F001, F003, F007)
18834968	<i>A connective tissue disorder caused by mutations of the LH3 gene</i>	First human case; recessive/compound-het; variant-specific loss of each enzymatic activity (F001, F004, F008)
16467571	<i>Glycosylation catalyzed by LH3 is essential for basement membranes</i>	GGT activity is the essential function; dose-dependent embryo survival (F002)
17873278	<i>Secretion and assembly of type IV and VI collagens depend on glycosylation...</i>	Mechanism: loss of Hyl-glycosylation blocks Col VI tetramerization / Col IV,VI secretion; Ullrich CMD muscle overlap (F002, F009)
37446392	<i>Regulatory "Hot Spots" for LH/ PLOD glycosyltransferase activity</i>	Enzymatic division of labor: LH3=Glc-T, GLT25D1=Gal-T (F002)
30089812	<i>Molecular architecture of LH3</i>	Homodimer, two catalytic sites; disease mutations near active sites (F004)
31129566	<i>Pathogenic variants in PLOD3...</i>	Clinical feature catalogue; Stickler/EDS/EB overlap (F003)
30463024	<i>PLOD3 mutations cause RDEB-like blistering...</i>	Skin phenotype; reduced glycosylated Hyl biomarker; reduced LH3 protein (F003, F006)
26380979	<i>LH3 localizes to epidermal basement membrane...</i>	LH3 BM localization/depletion; IHC diagnostic modality (F006)
36203519	<i>Cerebral small vessel disease caused by PLOD3...</i>	Severe neurovascular phenotype; homozygous p.Leu406del (F007)
27435297	<i>Regulation of post-Golgi LH3 trafficking...</i>	VPS33B/VIPAR sorting of LH3; ARC syndrome link (F005)

The remaining reviewed papers (e.g., PLOD3 in colorectal, lung, liver, bladder, esophageal cancers; PMIDs 39948137, 35265665, 35116582, 36872941, 41491166, 39659928, 34646265) concern PLOD3's oncologic/ECM-remodeling role and do **not** bear directly on BCARD pathophysiology; they are noted here only to document that LH3 also functions in tumor ECM stiffening — a context distinct from the germline loss-of-function disease.

Limitations and Knowledge Gaps

1. **Ultra-rarity (n ≈ 11):** All clinical conclusions rest on a handful of case reports. Frequencies, penetrance, expressivity, and prognosis cannot be quantified with confidence.
 2. **No epidemiological data:** Prevalence, incidence, carrier frequency, sex/ethnic distribution are unknown.
 3. **Genotype–phenotype correlation is immature:** Only a few variants are functionally characterized; whether specific alleles (e.g., GGT-selective vs LH-selective) predict organ-specific severity is unresolved.
 4. **Mechanistic inference for organ-specific features:** The link from collagen underglycosylation to bone, ear, and vessel phenotypes is strongly plausible but largely **inferred** from collagen biology rather than directly demonstrated in BCARD tissue.
 5. **No therapeutics or trials:** No disease-modifying therapy, no registered clinical trials, no validated QoL or prognostic instruments.
 6. **Model gaps:** Complete-knockout embryonic lethality limits study of postnatal features; conditional/hypomorphic, zebrafish, and iPSC/organoid models are underdeveloped or absent.
 7. **Biomarker validation:** Reduced glycosylated hydroxylysine is promising but not standardized as a clinical diagnostic assay across laboratories.
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Proposed Follow-up Experiments / Actions

1. **Establish an international BCARD/PLOD3 patient registry** to aggregate the scattered cases and generate the first real prevalence, penetrance, natural-history, and mortality data — with structured vascular-event and neurodevelopmental outcomes.

2. **Systematic genotype–phenotype study:** functionally classify each reported *PLOD3* variant for residual LH vs GGT activity (in vitro assays) and correlate with organ involvement and severity, testing the hypothesis that residual GGT activity predicts survival/severity (extending F002).
3. **Standardize the biochemical diagnostic:** validate a quantitative urinary/tissue glycosylated-hydroxylysine assay (ratio of glucosyl-galactosyl-Hyl to total Hyl) as a rapid, orthogonal confirmatory test alongside sequencing (F006).
4. **Develop conditional/hypomorphic mouse and human iPSC-derived models** (osteoblast, vascular smooth muscle, cochlear, cortical) to interrogate postnatal, tissue-specific mechanisms not accessible in embryonic-lethal knockouts (addressing F002/F007/F009 gaps).
5. **Vascular surveillance protocol:** define evidence-based imaging intervals and intervention thresholds for aneurysm/dissection, given rupture is the leading lethal risk (F003).
6. **Explore trafficking-based therapeutic hypotheses:** given the VPS33B/VIPAR–LH3 axis (F005), test whether enhancing LH3 stability/trafficking or supplying residual GGT activity rescues collagen glycosylation in patient fibroblasts — a first step toward any disease-modifying strategy.
7. **Cross-collagenopathy differential-diagnosis panel:** ensure *PLOD3* is included in Stickler/EDS/EB/Ullrich-CMD gene panels so that BCARD is not missed in patients presenting with overlapping features (F003, F009).

Report compiled from 9 confirmed findings and 25 reviewed papers over 5 investigation iterations. Evidence types: human clinical case reports (PMIDs 40289369, 31129566, 30463024, 36203519, 18834968, 26380979); in vitro/structural (30089812, 37446392, 26380979); mouse model (16467571, 17873278); trafficking/cell biology (27435297); population genomics (gnomAD, supporting F008).
